

CE 211 – Concrete Technology (Revised Cheat Sheet)

This document is a corrected and verified one-page cheat sheet based on the uploaded lecture "Course material_CE 211.pdf" (rechecked). It also lists clarifications and corrections I made to the original summaries I provided. Citations to the uploaded file are shown in the chat; this PDF omits those markers to stay compact. If you need the exact slide references, check the chat where I cite the file.

Ingredients (confirmed)

- Cement (OPC, PPC, PSC, blended, SRPC, RHPC)
- Water (potable) — controls hydration and workability; w/c controls strength
- Aggregates — coarse (gravel/crushed) + fine (sand) — grading, MSA, FM matter
- Admixtures: chemical (water-reducers, accelerators, retarders, air-entrainers)
- Mineral admixtures (SCMs): fly ash (Class F & C), GGBS, silica fume, rice husk ash

Cement chemistry & hydration

- Main clinker phases: C3S (early strength), C2S (late strength), C3A, C4AF
- Hydration (general): $C_3S/C_2S + H \rightarrow C-S-H$ (strength) + $Ca(OH)_2$ (portlandite)
- Pozzolanic reaction: $SCM + Ca(OH)_2 + H \rightarrow$ additional $C-S-H$ (slow but durable)
- Gypsum (~3–5% interground) controls C3A reaction and prevents flash set

Bogue equations (as in lecture)

- $C_3S = 4.07(CaO) - 7.60(SiO_2) - 6.72(Al_2O_3) - 1.43(Fe_2O_3) - 2.85(SO_3)$
- C_2S (commonly written) = $2.87(SiO_2) - 0.754(C_3S)$ (PDF shows an alternative notation)
- $C_3A = 2.65(Al_2O_3) - 1.69(Fe_2O_3)$; $C_4AF = 3.04(Fe_2O_3)$
- Note: Bogue gives theoretical phase estimates — actual clinker phases depend on kiln practice

Admixtures & SCMs (key points)

- Water reducers (plasticizers): ~5–15% water reduction; Superplasticizers (PCE/SNF/SMF): ~25–35%
- Accelerators: e.g., $CaCl_2$ (effective but chloride can corrode reinforcement) — avoid for reinforced concrete
- Retarders: delay set (useful in hot weather/long transport) — e.g., carbohydrate derivatives
- Air-entrainers: microscopic bubbles (improve freeze-thaw resistance); PDF lists bubble sizes 20–2000 μm
- Fly ash (Class F & C), GGBS, silica fume (0.1 μm particles) and RHA: benefits—improved later strength, reduced permeability, lower heat, economy

Workability & tests

- Workability = consistency (flow) + cohesiveness (resistance to segregation/bleeding)
- Tests (PDF cites IS 1199 Part 2): Slump, Vebe, Compacting factor
- Slump cone (standard dimensions used internationally): height 300 mm; base 200 mm; top 100 mm. (Follow IS 1199 for official procedure)
- Moisture correction: $Absorption\% = (W_{ssd} - W_{dry})/W_{dry} \times 100$; $Total\% = (W_{wet} - W_{dry})/W_{dry} \times 100$; $Surface = Total - Absorption$

Fresh concrete problems

- Segregation: coarse particle separation (dry mixes) or grout separation (wet mixes) — controlled by grading & handling
- Bleeding: water rises to surface → weak surface layer; control by lower w/c, air entrainment, proper placement

Corrections, Clarifications & Additions (I rechecked the uploaded PDF and my earlier summaries)

Bogue C2S formula (clarified)

The PDF shows a C2S expression that is written using a term ' $3\text{CaO} \cdot \text{SiO}_2$ '. The commonly used cleaned form is shown in this sheet: $\text{C}_2\text{S} \approx 2.87(\text{SiO}_2) - 0.754(\text{C}_3\text{S})$. I corrected my earlier shorthand to this clearer form.

Fineness (added detail)

I previously quoted a typical Blaine range; the lecture mentions fineness concept but not numeric ranges. Typical Blaine fineness for OPC is about 250–350 m^2/kg (varies by manufacturer). Use the cement manufacturer's datasheet for exact values.

Slump test details (procedure)

The PDF references IS 1199 but does not list the cone dimensions or rodding strokes. I added common standard details earlier (300 mm cone height; 25 rods per layer). These are standard practice, but ALWAYS follow IS 1199 exact steps in the exam setting if the question references the standard.

CaCl₂ caution (safety note)

The lecture lists CaCl₂ as a widely used accelerator. I added the widely accepted caution that chlorides accelerate reinforcement corrosion and should be avoided in reinforced concrete unless corrosion mitigation is assured.

Air entrainment strength effect (clarified)

I earlier gave a simple ' $\approx 5\%$ strength loss per 1% air' — that is a rule-of-thumb that varies. Use it as a rough estimate only; actual effect depends on aggregate, mix, and cement.

SSD & moisture formulas (confirmed)

The moisture formulas and SSD definition in the lecture are correct and are used in my moisture-correction example. See the uploaded PDF for the exact expressions.

Added, useful exam topics (not in PDF)

Important topics often tested but not detailed in the PDF: curing importance & typical durations (keep moist — commonly min. 7 days for OPC in many situations), standard compressive strength test references (IS 516/ASTM C39), alkali-silica reaction basics and mitigation (use of low-alkali cement and SCMs), and mass concrete thermal control (use low heat cements/SCMs). I can expand these on request.

Final Quick Checklist (for exam)

- Know ingredient roles + gypsum purpose (control C3A).
- Memorize hydration equations and what C–S–H and $\text{Ca}(\text{OH})_2$ are.
- Understand Bogue purpose and that it is theoretical (practice one calculation).
- Practice moisture correction (SSD vs wet mass) and one slump/compacting-factor interpretation.
- Remember admixture categories and their key effects (plasticizer vs superplasticizer; CaCl₂ caution; air entrainment tradeoffs).